A380X electrical alerts and AC ESS loads

The A380X ECAM raises the FCOM ELEC alerts from the electrical network, the AC ESS loads are on the AC ESS busbar, the two transponders and the second NSS ANSU are wired, and new battery, EXT PWR, RAT generator, TR monitoring and drive disconnect failures come with their ELEC alerts

Aircraft	A380X
Where to find it	Overhead ELEC panel; flyPad > Failures > ATA 24; E/WD and SD (ELEC AC / DC, STATUS); pedestal SURV panel and MFD SURV > STATUS & SWITCHING; OIT
Branch	`core/a380x-elec`, based on `core/gps` (stacked: needs core/surveillance, core/a380x-mfd-fms and core/gps merged first).
Status	Tested in the sim by the user: ELEC AC BUS 2 and AC BUS 4 FAULT OK; ELEC AC ESS BUS FAULT with its STATUS, GEN OFF, the AC ESS and AC EMER failures, XPDR 1 / XPDR 2 with the SURV alerts and ANSU 1 / 2 OK. Battery and EXT PWR failures (pick): OK on a combined test build; the RAT generator, TR monitoring and drive disconnect failures are not sim-tested (unit and Rust tests). The other ELEC alerts are not sim-tested (unit tests: 30). A full MSFS restart is needed after installing (systems WASM, cockpit XML).
Pictures	No pictures: the alerts use the stock E/WD and SD STATUS pages.

1. ELEC alerts from the electrical network
2. AC ESS and AC EMER loads
3. XPDR 1 / XPDR 2, per-system TCAS faults and SURV XPDR alerts
4. NSS AVNCS ANSU 2
5. Battery, EXT PWR, RAT generator, TR monitoring and drive disconnect failures

1. ELEC alerts from the electrical network

> **Aircraft:** A380X | **Where:** E/WD; SD ELEC AC / DC and STATUS | **Status:** Tested in the sim by the user: AC BUS 2 / AC BUS 4 FAULT OK, AC ESS BUS FAULT with STATUS and GEN OFF OK. Other alerts not sim-tested (unit tests: 30).

What it is for

The A380X ECAM had the ELEC alert texts but nothing raised them. The flight warning system now computes them from the electrical network (busbars, engine and APU generators, TRs, pushbuttons), with the procedures, the flight phase inhibitions and the master lights of A380 FCOM PRO-ABN-ECAM-10-24:

- **ELEC AC BUS 1(2)(3)(4) FAULT**, **AC ESS BUS FAULT**, **AC ESS BUS ALTN** (AC ESS supplied through AC 4: AC ESS FEED in ALTN), **DC BUS 1(2) FAULT**, **DC ESS BUS FAULT**, **EMER CONFIG** (all four AC busbars lost, master warning).
- **ELEC GEN 1(2)(3)(4) FAULT** (stays, with its GEN OFF line ticked, after the generator is switched off, until it works again), **GEN 1(2)(3)(4) OFF** (GEN pb-sw OFF with its engine running, after 5 s), **DRIVE 1(2)(3)(4) DISCONNECTED**, **APU GEN A(B) FAULT**, **TR 1(2)(ESS) FAULT** (no master light, no chime), **APU TR FAULT**, **BUS TIE OFF**.
- Procedure lines are sensed where the FCOM says so (switches set, OFF THEN ON resets, AIR DATA SWTG, WXR & TAWS / XPDR & TCAS SYS 2...). Lines that the FBW texts had from another FCOM standard and that the FCOM procedure does not contain are hidden.
- ELEC AC BUS 2 FAULT shows **LDG DIST: AFFECTED** (FCOM: LDG DIST IMPACT ON WET/CONTAM RWY ONLY).
- **ELEC AC ESS BUS FAULT** has its FCOM STATUS: LIMITATIONS (APPR & LDG) SLATS SLOW; INOP SYS all phases ADR 1, GPS 1, WXR 1, XPDR 1, TCAS 1, HF 1, FEED TK 2 MAIN PMP, STBY PITOT HEATG, L WINDSHIELD HEATG, DFDR; INOP SYS APPR & LDG SLAT SYS 1, CAT 2, GLS AUTOLAND, LS 1, TAWS 1; REDUND LOSS (STATUS MORE) RA SYS C, FEED TK 3 STBY PMP, TRIM TK L PMP, ENG 1+2+3+4 IGN A, PACK 1 CTL 1, PACK 2 CTL 1. GPWS 1, TERR SYS 1 and ADS-B RPTG 1 are not shown as separate alerts while it is active.

> **NOTE**
> The XPDR 1 INOP SYS text is defined once for the ELEC and the SURV XPDR alerts, and the L WINDSHIELD HEATG INOP SYS text of this STATUS is added here (the same text and code as on `core/ecam-fws`).

Limitations and why

Limitation	Why	What to do
A bus fault alert needs at least one AC busbar powered.	Design choice: with all four AC busbars off the aircraft is unpowered on the ground or in ELEC EMER CONFIG, whose procedure covers the lost busbars.	None.
"Failed" means "no normal output although it should give one".	The sim gives no health word for the generators and TRs, only their output.	None.
Not done here: combined busbar faults, DRIVE OIL, units that are not simulated, T.O CONFIG.	First part of the ELEC alerts. The battery, EXT PWR, RAT, TR monitoring and drive disconnect alerts and the STATUS of the GEN, DRIVE, APU GEN and TR alerts came later (next feature); the combined busbar alerts and the T.O CONFIG check of the GEN / DRIVE alerts are on `core/a380x-elec-fuel-alerts`.	Use that branch for them.
VENT AVNCS EXTRACT lines are never ticked.	There is no avionics ventilation pushbutton in the sim.	None.
REDUND LOSS needs the ECP MORE key fix.	The MORE key and light fix is on `core/ecam-fws`.	Merge that branch to page to STATUS MORE.

How to use

1. When an ELEC alert shows, follow the E/WD procedure (the sensed lines tick by themselves as you act).
2. Check the ELEC AC / DC SD page (called with the alert) and the STATUS page afterwards.

How to test

#	Do this	Expected result
1	A380X on the ground, four engines running, all GEN pb-sw ON, BUS TIE AUTO. flyPad > Failures > ATA 24 Electrical Power > **AC 2**.	E/WD: **ELEC AC BUS 2 FAULT** with its procedure, including LDG DIST: AFFECTED; SD ELEC AC page.
2	Clear it. Fail **AC 4**.	**ELEC AC BUS 4 FAULT**; on the ground the line FOR ENG STOP : KEEP ENG 1 LAST.
3	Clear it. Fail **AC ESS** (this label is the A380 AC ESS busbar).	**ELEC AC ESS BUS FAULT**, first line AC ESS FEED ..... ALTN; FAULT light in the AC ESS FEED pb-sw. No separate GPWS 1, TERR SYS 1 or ADS-B RPTG 1 alert.
4	Press CLR until the STATUS page shows (or press STS).	SLATS SLOW; INOP SYS ADR 1, GPS 1, WXR 1, XPDR 1, TCAS 1, HF 1, FEED TK 2 MAIN PMP, STBY PITOT HEATG, L WINDSHIELD HEATG, DFDR; APPR & LDG SLAT SYS 1, CAT 2, GLS AUTOLAND, LS 1, TAWS 1. With `core/ecam-fws`: MORE shows the REDUND LOSS list.
5	Clear it. Set the **GEN 2** pb-sw OFF (engine 2 running).	After 5 s: **ELEC GEN 2 OFF**. GEN 2 ON: the alert goes.

6	Fail **Generator 3**.	**ELEC GEN 3 FAULT** with GEN 3 ... OFF. Set GEN 3 OFF: the line ticks and the alert stays. Clear the failure and set GEN 3 ON: the alert goes.
7	APU running, APU GEN A pb-sw ON. Fail **APU Generator 1**.	**ELEC APU GEN A FAULT** with APU GEN A ... OFF.
8	Fail **TR 1**, then **TR APU**.	**ELEC TR 1 FAULT** (no master caution, no chime), then **ELEC APU TR FAULT**.
9	Fail **DC 1**, then (cleared) **DC ESS**.	**ELEC DC BUS 1 FAULT**, then **ELEC DC ESS BUS FAULT**, with their procedures.
10	No failure. Set the BUS TIE pb-sw OFF.	**ELEC BUS TIE OFF**.
11	Set the AC ESS FEED pb-sw to ALTN.	**ELEC AC ESS BUS ALTN**.
12	In flight, fail **AC 1**, **AC 2**, **AC 3** and **AC 4**.	**ELEC EMER CONFIG** in red (master warning) with its procedure; the descent / APU lines depend on whether it triggered above or below FL 230.

2. AC ESS and AC EMER loads

> **Aircraft:** A380X | **Where:** Overhead ELEC panel (AC ESS FEED); flyPad failures AC ESS / AC EMER | **Status:** Tested in the sim by the user: OK with the AC ESS and AC EMER failures.

What it is for

The A380X electrical model uses the A320 bus names: the real A380 **AC ESS** busbar (400XP) is the A320 "AC ESS SHED" bus in the code, and the **AC EMER** busbar (491XP) is the A320 "AC ESS" bus. About twenty consumers read the wrong one, so they were on AC EMER where the FCOM puts them on AC ESS. They now use the AC ESS busbar (TypeScript, cockpit XML and Rust):

- SURV SYS 1 (XPDR 1, TCAS 1, TAWS 1, WXR 1), RA SYS C, the slat motor, the 427XP EHAs, the feed tank 2 main, feed tank 3 standby and trim tank left pumps, channel 1 of both pack controllers, the F/O OIT and the FLT OPS ANSU (A380 FCOM DSC-24, DSC-34-20-100, DSC-46-20-70, DSC-21).
- The **AC ESS FEED** FAULT light shows a loss of the AC ESS busbar (FCOM DSC-24-20: "The AC ESS busbar is not supplied"). An AC EMER failure leaves AC ESS supplied: no FAULT light.
- The manual crossbleed control is powered by DC ESS (FCOM DSC-36-30) and the outflow valve manual control by its controller supplies DC 1 / 2 or DC ESS (FCOM DSC-21-30-10). Before, both were on A320 buses that the A380 does not have, so they had no power.
- With only the batteries (static inverter on AC EMER), the AC ESS loads (427XP EHAs, TAWS 1, XPDR 1...) are unpowered and the AC EMER loads (VOR 1, LS 1, GPS 1...) remain.

Limitations and why

Limitation	Why	What to do
The flyPad labels are AC ESS (the real AC ESS busbar) and AC EMER; the code names stay the A320 ones.	Renaming the buses in the code would touch every system.	Use the flyPad labels.

How to use

1. Nothing new to set. In an AC ESS loss, follow ELEC AC ESS BUS FAULT (AC ESS FEED ALTN).

How to test

#	Do this	Expected result
1	A380X powered, engines running. flyPad > Failures > ATA 24 Electrical Power > **AC ESS**.	AC ESS FEED pb-sw FAULT light on; ELEC AC ESS BUS FAULT; the F/O OIT goes dark; MFD SURV STATUS & SWITCHING shows XPDR 1 and TCAS 1 failed.
2	Clear it. Fail **AC EMER**.	No AC ESS FEED FAULT light, no ELEC AC ESS BUS FAULT; XPDR 1, TCAS 1 and the F/O OIT keep working.

3. XPDR 1 / XPDR 2, per-system TCAS faults and SURV XPDR alerts

> **Aircraft:** A380X | **Where:** Pedestal SURV panel; MFD SURV > STATUS & SWITCHING; E/WD; SD STATUS | **Status:** Tested in the sim by the user: OK with XPDR 1, XPDR 2 and both failed.

What it is for

The A380 has two surveillance systems, each with its transponder and TCAS: SURV SYS 1 on AC ESS and SURV SYS 2 on AC 4 (FCOM DSC-34-20-100). MSFS has one transponder: it now acts as the transponder of the selected system (XPDR & TCAS SYS 1 / SYS 2 on the SURV panel or the MFD SURV page). The XPDR 2 failure did nothing before.

- XPDR 1 failed or unpowered: TCAS 1 is inoperative too; same for 2. The sim TCAS computer fault counts for the selected system only.
- **SURV TCAS 1(2) FAULT** is per system: no false TCAS 2 FAULT when AC ESS is lost (TCAS 2 is on AC 4). They need the busbar of that system powered (ELEC AC ESS BUS FAULT lists XPDR 1 / TCAS 1 itself).
- New **SURV XPDR 1 FAULT**, **XPDR 2 FAULT** and **XPDR 1+2 FAULT** (FCOM PRO-ABN-ECAM-10-34-20): caution, line XPDR & TCAS ..... SYS 2 (1) when the failed system is selected and the other one works; STATUS INOP SYS XPDR 1(2) and TCAS 1(2), plus AP/FD TCAS MODE when the failed system is selected; XPDR 1+2 FAULT: XPDR 1+2, TCAS 1+2. The TCAS FAULT alert of the same system is not shown with it.
- MFD SURV STATUS & SWITCHING: XPDR 1(2) and TCAS 1(2) shown failed when that XPDR is failed or unpowered.

How to use

1. Select the XPDR & TCAS system on the pedestal SURV panel or MFD SURV > STATUS & SWITCHING.
2. On SURV XPDR 1(2) FAULT, select the other system as the E/WD line asks.

How to test

#	Do this	Expected result
1	A380X on the ground, powered, XPDR & TCAS SYS 1 selected. flyPad > Failures > ATA 34 Navigation > **XPDR 1**.	E/WD: **SURV XPDR 1 FAULT** with XPDR & TCAS ..... SYS 2. MFD SURV STATUS & SWITCHING: XPDR 1 and TCAS 1 failed. STATUS: INOP SYS XPDR 1, TCAS 1, AP/FD TCAS MODE. No separate TCAS 1 FAULT.
2	Select SYS 2.	The line ticks; AP/FD TCAS MODE leaves the STATUS; the transponder works again (as XPDR 2).
3	Also fail **XPDR 2**.	**SURV XPDR 1+2 FAULT**; STATUS: XPDR 1+2, TCAS 1+2.
4	Clear both; with SYS 1 selected fail only **XPDR 2**.	**SURV XPDR 2 FAULT** without the switching line; STATUS XPDR 2, TCAS 2, no AP/FD TCAS MODE.
5	Clear it. flyPad > Failures > ATA 24 Electrical Power > **AC ESS**.	No SURV TCAS 2 FAULT (before: a false one); XPDR 1 / TCAS 1 are listed by ELEC AC ESS BUS FAULT.

4. NSS AVNCS ANSU 2

> **Aircraft:** A380X | **Where:** OIT (NSS AVNCS applications) | **Status:** Tested in the sim by the user: OK.

What it is for

A380 FCOM DSC-46-20-30: the avionics side of the network server system (NSS AVNCS) has two servers; ANSU 2 provides the functions when ANSU 1 fails. Only ANSU 1 existed (the ANSU 2 failure did nothing). ANSU 2 is now built and takes over; with both failed the NSS AVNCS applications (AOC, eLogbook, maintenance) are lost. Supplies (FCOM DSC-46-20-70): ANSU 1 AC 2 or BAT 2, ANSU 2 AC 1 or AC EMER or BAT 1, FLT OPS ANSU AC ESS or BAT 1.

How to use

1. Nothing to set: the OIT avionics applications use whichever NSS AVNCS ANSU works.

How to test

#	Do this	Expected result
1	A380X powered, OIT on an NSS AVNCS page. flyPad > Failures > ATA 46 Information Systems > **NSS AVNCS ANSU 1**.	The OIT avionics applications stay available (ANSU 2).
2	Also fail **NSS AVNCS ANSU 2**.	The NSS AVNCS applications are lost on the OIT.
3	Clear both failures.	The applications become available again.

5. Battery, EXT PWR, RAT generator, TR monitoring and drive disconnect failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 24 Electrical Power; overhead ELEC panel (BAT, EXT PWR, DRIVE pbs); E/WD, STATUS | **Status:** Tested in the sim by the user on a combined test build: BAT 1 and EXT PWR 1 OK. The RAT generator, TR monitoring and drive disconnect failures and the new STATUS lines: not sim-tested. Unit tests FwsElecSystemAlerts.spec.ts, a380ElecFailures.spec.ts; Rust tests. The core branch itself was not sim-tested.

What it is for

Sixteen new flyPad failures, simulated in the Rust electrical system, with the ELEC alerts of A380 FCOM PRO-ABN-ECAM-10-24:

flyPad failure (ATA 24)	What it does	ECAM
BAT 1, BAT 2, BAT ESS, APU BAT	The battery disconnects from its DC busbar; FAULT light in its BAT pb-sw (in either position, while failed).	ELEC BAT 1 (2)(ESS) FAULT or APU BAT FAULT, line BAT ... OFF; master caution and single chime; STATUS INOP SYS BAT 1 (2) (ESS), BAT 1+2 with both, APU BAT (INFO APU BAT START NOT AVAIL with the APU TR failed).
EXT PWR 1, 2, 3, 4	The GGPCU of that external power keeps its contactors open: no external power on that side.	ELEC EXT PWR 1 (2)(3)(4) FAULT while the ground power unit is connected, line EXT PWR n P/B RECYCLE (done after OFF then ON); no master light, no chime; INOP SYS EXT PWR n.
RAT Generator	The RAT generator gives no output: the RAT does not supply the AC ESS busbar. Detected RAT stowed or deployed.	ELEC RAT FAULT (crew awareness, no master light, no chime); INOP SYS RAT.
TR 1 Monitoring, TR 2 Monitoring, ESS TR Monitoring	The monitoring function fails; the TR keeps working.	ELEC TR 1 (2)(ESS) MONITORING FAULT (crew awareness, no master light, no chime, no STATUS).

DRIVE 1 (2)(3)(4) Disconnect Function	The DRIVE pb no longer disconnects that generator drive.	ELEC DRIVE 1 (2)(3)(4) DISC FAULT (master caution, single chime, crew awareness, no STATUS).

Also in this feature

- **ELEC BAT 1 (2)(ESS)(APU) OFF**: a battery pb-sw abnormally OFF for 5 s (line BAT ... ON). Not for a failed battery (its procedure asks OFF), not for the APU BAT while ELEC APU TR FAULT asks it OFF, and not with all four batteries OFF together (aircraft secured, battery check).
- The GEN, DRIVE, APU GEN and TR alerts now list their STATUS: INOP SYS GEN n, APU GEN A (B), TR 1 (2)(ESS), APU TR (INFO APU START NOT AVAIL with the APU BAT failed); for a lost engine generator (GEN n FAULT, GEN n OFF, DRIVE n DISCONNECTED) with the APU off: APPR & LDG INOP SYS CAT 3 DUAL and INFO CAT 3 SINGLE ONLY.
- Flight phase inhibitions of the FCOM figures: BAT FAULT / OFF phases 3 to 7, 9 and 10; EXT PWR FAULT 3 to 10; RAT FAULT 3 to 11; TR MONITORING FAULT and DRIVE DISC FAULT 2 to 11.

Limitations and why

Limitation	Why	What to do
ELEC RAT FAULT covers only the RAT generator.	There is no RAT uplock or heater in the simulation.	-
ELEC EXT PWR FAULT shows only with the ground power unit connected.	Design choice: without one there is nothing to supply.	Connect ground power first.
The TR monitoring alerts read the flyPad failures directly.	The failure changes nothing in the network (design choice).	-
A full MSFS restart is needed after an update.	The systems WASM module changes.	Restart the sim after installing.

How to use

1. Open flyPad > Failures > ATA 24 Electrical Power and activate one of the new failures.
2. Follow the E/WD (BAT ... OFF, EXT PWR ... RECYCLE) and read the STATUS page.
3. Deactivate the failure and set the pb-sw back.

How to test

Setup: cold and dark at the gate, all four BAT pbs AUTO, EXT 1 ON, APU off.

#	Do this	Expected result
1	flyPad > Failures > ATA 24 Electrical Power > **BAT 1**.	ELEC BAT 1 FAULT (amber, single chime), FAULT light in the BAT 1 pb-sw, line BAT 1 ... OFF.
2	Overhead ELEC panel: BAT 1 pb-sw OFF.	The line ticks; no ELEC BAT 1 OFF caution.
3	Deactivate BAT 1; BAT 1 pb-sw AUTO. Start the APU, APU GEN A and B ON. flyPad > Failures > ATA 24 Electrical Power > **EXT PWR 1**.	ELEC EXT PWR 1 FAULT (no chime); AC 1 is taken over by the APU.
4	EXT 1 pb OFF, then ON. Deactivate EXT PWR 1.	The EXT PWR 1 P/B RECYCLE line ticks after OFF / ON; the alert goes.
5	No failure, engines off, batteries AUTO: set the BAT 2 pb-sw OFF, wait 5 s.	ELEC BAT 2 OFF with BAT 2 ... ON. BAT 2 AUTO: the alert goes.

6	Still on the ground with the engines off (flight phase 1: these alerts are inhibited from phase 2). flyPad > Failures > ATA 24 Electrical Power > **DRIVE 3 Disconnect Function**.	ELEC DRIVE 3 DISC FAULT (master caution, single chime). Clear it.
7	flyPad > Failures > ATA 24 Electrical Power > **TR 1 Monitoring**, then **RAT Generator**.	ELEC TR 1 MONITORING FAULT and ELEC RAT FAULT, no master caution, no chime; STATUS INOP SYS RAT. Clear both.
8	Engines running, APU off. Fail **Generator 3**.	ELEC GEN 3 FAULT; STATUS: INOP SYS GEN 3, APPR & LDG CAT 3 DUAL, INFO CAT 3 SINGLE ONLY.